



Society of Petroleum Engineers

SPE-227813-MS

Improvements in Drilling Performance and Reduced Tool Loads in Challenging Vibration Prone Applications with Solutions from Lab and Field Testing

D. Daksuev, Baker Hughes, Drill Bits, Dhahran, Kingdom of Saudi-Arabia; A. Kueck, Baker Hughes, Drill Bits, The Woodlands, USA; F. Valbuena, Baker Hughes, Drill Bits, Dhahran, Kingdom of Saudi-Arabia; M. Savage, J. Quintero, and J. Bomidi, Baker Hughes, Drill Bits, The Woodlands, USA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227813-MS](https://doi.org/10.2118/227813-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The oil and gas industry pushes for longer, faster, and more reliable drilling operations. Predicting and optimizing drilling performance offers tremendous potential to achieve high rate-of-penetrations. Torsional and lateral vibrations are detrimental to bits and drilling tools. Specifically, High-Frequency Torsional Oscillations (HFTO) can lead to electronic failures, body cracks and twist-offs of drilling tools. A recently upgraded full-scale drilling rig enables the analysis of drilling speed and generated vibrations under laboratory conditions. This paper presents methods to generate heat maps to optimize drilling parameters to increase ROP while simultaneously mitigating harmful vibration dysfunctions. The method is validated by full-scale laboratory tests and field operations.

Three bit designs were used to drill rocks under realistic downhole pressure and weight-on-bit (WOB) and revolutions-per-minute (RPMs) in the lab. High-frequency sensors at the rig and in the bit capture the rate-of-penetration and the dynamic response at multiple combinations of operative parameters. The data allow a full assessment of the performance, efficiency and lateral and torsional stability of bits using stability-maps, rate-of-penetration-maps (ROP-maps), mechanical-specific-energy-maps (MSE maps) and depth-of-cut-weight-on-bit (DOC-WOB) curves. The lab tests are supported by 3D full bit simulations. The lab results are compared to commercial field tests in vibration prone rocks the middle east. The field runs were drilled in comparable well paths, formations and BHAs enabling a direct comparison. The bottom-hole-assemblies (BHA) were simulated and compared to high-frequency downhole data, surface data and offset-wells. The new procedure covers a wider range of parameters and the resulting stability maps are used to deduce recommendations for choice and operation of the drill bits to reduce loads on the BHA while increasing drilling performance.

The laboratory tests identified differences in ROP, torsional and lateral stability between the tested bit designs. Stability maps revealed stable regions of RPM-WOB combinations free of torsional vibrations. HFTO can be mitigated by increasing the rotational speed above an RPM threshold. Applying the new operative procedure in field operations shows that the stability maps are very comparable to the lab results

despite differences in the BHA setup. Learnings generated in the laboratory are translatable to improvements in the field. Certain rock types provide less vibration excitation leading to a larger vibration free-zone identified through the stability maps.

The new testing procedure and evaluation method enables to improve bit designs, develop operational recommendations quicker, minimize costs, and deliver more precise and reliable solutions. Improvements of the performance and the torsional stability simultaneously are made possible through the new testing methods. The suppression of HFTO by optimized bit and BHA design combined with expanded stable operating parameters will lead to increased tool reliability, less NPT and higher drilling performance.

Introduction

Longer, faster and more reliable drilling operations are critical aspects of well construction economics that ultimately drives increased production. Torsional and lateral vibrations are detrimental to bits and drilling tools. Specifically, High-Frequency Torsional Oscillations (HFTO) can lead to electronic failures, body cracks and twist-offs of drilling tools. Predicting and optimizing drilling performance offers tremendous potential to achieve high rate-of-penetrations. From Tool selection standpoint, this study's primary focus is on the drill bit improvements that can be done as an addition to the BHA improvements that further drive drilling performance. Methods described here are towards generating heat maps to optimize drilling parameters to increase ROP while simultaneously mitigating harmful vibration dysfunctions.

Application Background

The 8-1/2" build-up section is usually drilled to land the well two feet TVD above the target reservoir with motorized rotary steerable systems. The section's geologic setting includes several key formations, each of which presents unique drilling challenges due to its heterogeneous nature and varying mechanical properties. This application is most common in the region and was chosen due to consistent drilling vibrations problems that lead to downhole tool fatigue and failures despite different BHA optimization. The current project aims to reduce harmful vibrations bringing significant improvement in RSS tool reliability, decrease risks of NPT and can improve ROP.



Figure 1—The Middle East as one of the focus regions of this study

Formations

The build-up section consists of four main formations (I through IV), each of which presents its own set of drilling complexities due to variations in both composition and mechanical properties.

Formation (I). This formation is a cross-bedded, medium-grained sandstone interbedded with minor siltstone and shale. The primary challenges associated with this formation have been wellbore instability, characterized by potential collapse and caving. Recent geomechanical studies in the region have shown that the instability of this formation is primarily due to its highly stressed state and mineralogical composition.

Formation (II). This Upper Tithonian formation is a massive limestone formation, more than 100 meters thick, that marks the return to an open marine environment after the deposition of the Late Jurassic anhydrite evaporites in the interior of the Arabian Plate.

Formation (III). This formation is characterized by its predominant anhydrite composition interbedded with limestone and carbonate beds, typically exceeding 90 meters in thickness. It consists of three distinct members: a lower anhydrite-dominated section, a middle mixed zone of anhydrite and carbonates, and an upper carbonate unit. The formation displays a variety of textures including chicken wire, nodular and laminated structures, along with internal deformations such as brecciated shear zones and minor faults. Formed in a shallow subaqueous environment, it serves as an essential regional seal for hydrocarbon reservoirs, with carbonate content typically less than 20%.

Formation (IV). This formation is a significant Late Jurassic carbonate-evaporite sequence characterized by alternating carbonate and anhydrite beds. It consists of four main reservoir units (A, B, C, and D), with dolomite as the primary lithology and anhydrite as the dominant evaporite component. The formation was deposited on a carbonate ramp under arid conditions, with eight distinct sedimentary facies.

Formations II and III were focused on in this study since both are main source of vibrations such as stick-slip, high-frequency torsional oscillations (HFTO) and lateral vibrations.

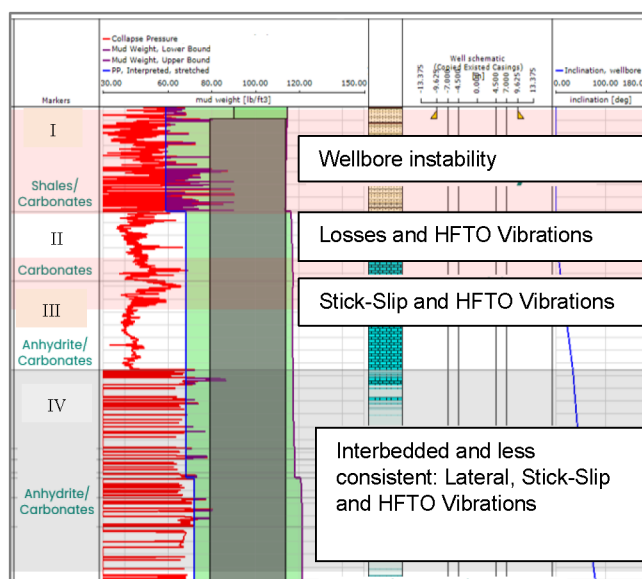


Figure 2—Example formation layers of one of the applications

Vibrations and dynamic dysfunctions

While drilling formations I through IV, significant amount of vibrations are experienced by the drill bit and the drilling tools that may affect the drilling performance and the tool life (Reckmann et al. 2008, Kueck et al. 2017). They can be classified into axial, lateral, and tangential vibrations, see Fig 3 (Jansen 1991, Heijden 1994, Dykstra 1996, Germay et al. 2009, Ostermeyer & Kueck 2013). Each direction is associated

to a specific dynamic dysfunction acting in a certain frequency range. The sensors in drilling tools such as MWDs, steering units and in-bit sensing devices, require a bandwidth high enough to detect such vibrations otherwise data might be misinterpreted.

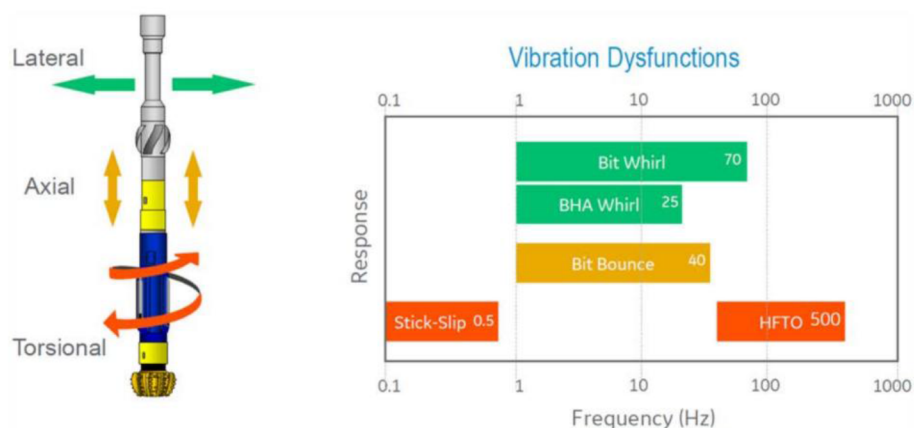


Figure 3—Acceleration modes (left) with matching dysfunction types and frequency ranges (right)

Lateral Vibrations. Lateral vibrations negatively affect drilling performance by damaging the cutting structure (Brett et al., 1989, Warren et al. 1990) and by reducing hole quality, which in turn impacts steerability. Two primary lateral vibration modes—forward and backward whirl—are illustrated in Fig. 4. Backward whirl is characterized by the bit's geometric center rotating around the borehole center in the opposite direction of bit rotation. This high-frequency motion of up to 70 Hz generates high lateral accelerations and overloads the cutters, often resulting in chipping or delamination. In contrast, forward whirl is a lower-frequency (1 - 5Hz) motion where one of the blades is sliding along the borehole wall and bit's center is rotation around the hole center in the same sense of rotation. Forward whirl enlarges the hole and leads to asymmetric bit blade wear. The likelihood and severity of backward whirling of the drill bit are determined by the RPM, the ratio of borehole to bit radius (Jansen, 1993), the bit mass imbalance, the bit force imbalance and high tangential forces at the bit shoulder. It occurs at low depth-of-cut (DOC) and high RPM, where cutter engagement is low and imbalance forces are high. Lateral vibrations interact with torsional vibrations as shown recently (Hohl et al. 2024).

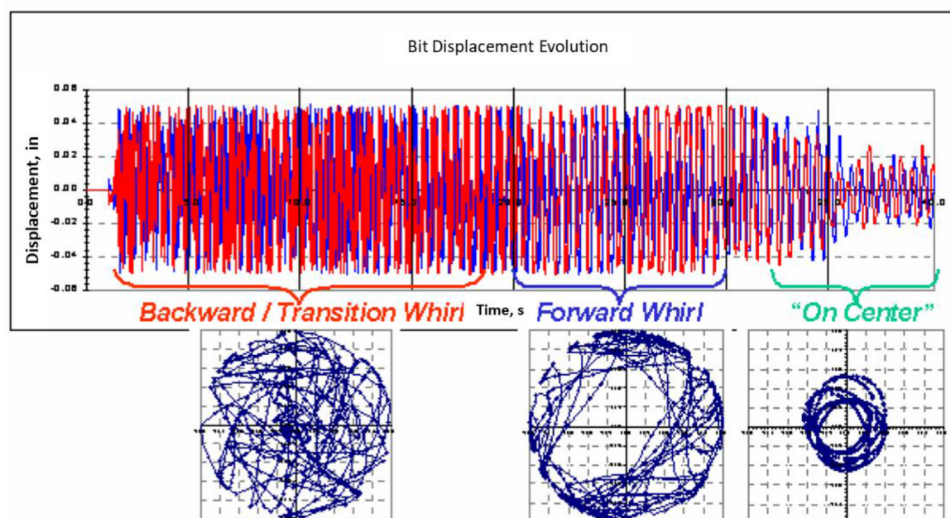


Figure 4—Lateral instability drilling patterns transitioning from backward to forward whirl to centric drilling

Torsional Vibrations including High-Frequency Torsional Oscillations. Destructive loads experienced by the BHA may be generated by High-Frequency Torsional Oscillations (HFTO, Sugiura et. al. 2020, Zhang et. al. 2023, Wilson et al 2024). This type of torsional vibration was initially observed through high tangential accelerations of high amplitude (Pastusek et al. 2007, Lines et al. 2013). Further analyses revealed HFTO is a self-excited torsional vibration mode localized in the lower section of the drillstring (Oueslati et al. 2013). Different from the well-known low frequency torsional phenomenon of stick-slip, HFTO is typically excited at a high frequency range of 50 Hz – 500 Hz (Jain et al. 2014, Fu et al. 2022, Sugiura et al. 2019). The excited frequency is determined by the geometry and the material of the lower drill-string, the BHA, and the bit. Like stick-slip, HFTO is likely to be excited at low RPM and high WOB (Hohl et al. 2020a, Hohl et al. 2024).

The excitation of HFTO is attributed to the bit-rock interaction during the engagement of cutting elements of polycrystalline diamond cutter (PDC) bits in certain rock types (Jain et al. 2014, Fu et al. 2022, Fu et al. 2023). Typically, HFTO trigger only one single excited frequency at a time which is a torsional natural frequency of the drill-string.

The worst-case amplitude of HFTO can be determined by assuming the lower boundary of the static RPM to be zero (see Fig 5, also Hohl et al. 2015). Once HFTO is fully developed the dynamic amplitude $RPM_{dynamic}$ reaches a stable value. The stable dynamic amplitude is two times the static RPM_{static} ranging from 0 to $2 RPM_{static}$. The HFTO excitation follows an exponential increase and takes ~ 0.5 s to fully develop. This can also be described as negative damping (Kulke et al. 2019, Hohl et al. 2015). Theoretical and field case studies (Hohl et al. 2020b, Heinisch et al. 2020) have proven that stick-slip and HFTO can be superimposed. In this situation, stick-slip can cause the static RPM to increase by a factor of more than two (Hohl et al. 2018, Lam et al. 2020). Consequently, the HFTO amplitude will increase proportionally.

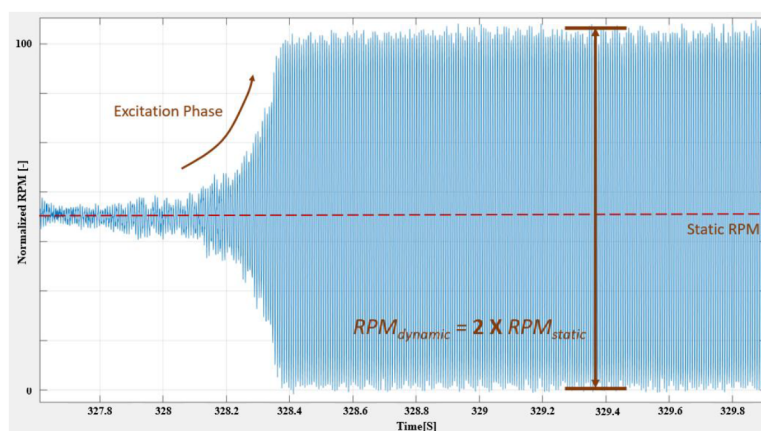


Figure 5—Excitation of HFTO in a laboratory downhole simulator drilling test, recorded by an in-bit sensing device

Tool failures

Excessive dynamic torque at high frequency presents risk of tool twist-off. Typical cracks due to extreme dynamic torque are observed at an angle of 45° (Nair et al. 2025). High tangential accelerations are also crucial to monitor since they can lead to electronic failures and cable damage. Electronics components sometimes experience loads above 100 g for extended time. The pictures below illustrate typical tool failures due to high dynamic torque caused by high stick-slip and high frequency torsional oscillations. For example, HFTO with a frequency of 200 Hz generate 200 load cycles per second at high dynamic torque and high accelerations. If these vibrations continue for extended time it can lead to material fatigue and potential tool failures, see Fig 6.



Figure 6—Examples for the impact of high-frequency tangential vibrations.

Goals

The goal of the lab and field-testing campaign is to reduce the number of drilling tool failures and increase NPT by reducing lateral and torsional loads on the tools generated by vibrations. Simultaneously, key performance indicators like ROP, steerability and durability must match the applications requirements as well. Solutions to achieve the goals involve identifying stability zones of operative parameters without vibrations to mitigate through adjustment of the drilling parameters and designing new bits and simulating and validating them in lab and field tests.

Explored Bit Designs

Three PDC bit designs were utilized during these trials, all in 8-1/2" bit diameters. The differences in the bit designs represented varying levels of design changes compared to the baseline frame, which had extensive history in the target application. The baseline design consisted of a 5 bladed arrangement comprising 16mm cutters with differing cutter types used throughout the bit. The bit's cutter profile and backrake scheme were developed based on learnings from previous iterations in the application. The bit body was constructed of tungsten carbide matrix, contained backup cutters on all blades, and utilized 7 SP nozzles (Alsaif et al. 2024). A second design concept utilized the same crown design as the baseline, but with a revised arrangement of cutters. A third design concept represented a greater design departure from the first two concepts. It also used 16mm cutters consisting of various cutter types, while adding a sixth blade to the cutting structure for additional cutter volume. Design features such as the bit body construction utilizing matrix were retained in the third concept. Isometric views of each design can be seen in Fig. 7.



Figure 7—Bit "A" on left. Bit "B" at center. Bit "C" at far right.

Modeling and simulation

BHA influence

The proprietary software TOA, based on finite element analysis, is used to simulate torsional vibrations. It models natural frequencies and their respective mode shapes, which can potentially be excited during bit-rock interaction. TOA simulates twisting motion that causes varying levels of torque in the drill string, focusing primarily on stick-slip and high-frequency torsional oscillation (HFTO).

Stick-slip is defined as the non-uniform rotation of the drill string caused by resistance to motion. The bit stops rotating ("stick"), which builds torque in the drill string. Once the torque reaches a critical point, it suddenly releases ("slip"), causing the bit and drill string to unwind rapidly. This typically occurs in the frequency range of 0–5 Hz.

HFTO, on the other hand, is a self-excited torsional vibration of the BHA/drill string caused by the interaction between the drill bit and the rock. Unlike stick-slip, which occurs at the first torsional eigenfrequency, HFTO occurs at higher natural torsional frequencies—typically between 50 and 500 Hz—and is concentrated in the BHA. HFTO induces cyclic fatigue loading on bits and drilling tools within the BHA, limiting tool life and drilling performance.

The results of actual BHA modeling (see Fig 8) show that dynamic torque can exceed bit torque by several times. High tangential accelerations were also observed, potentially surpassing electronic limitations depending on the shape of the triggered vibration mode. Stress analysis indicates that dynamic torque and acceleration are directly proportional to the maximum RPM for any given frequency. At extreme bit RPM values, this dynamic torque can accelerate fatigue in the mechanical components of the rotary steerable system (RSS). PDC bit design also influences the modeled frequency and dynamic torque. This study focuses on how PDC bit design can minimize the occurrence of high-frequency oscillations (see Fig. 8).

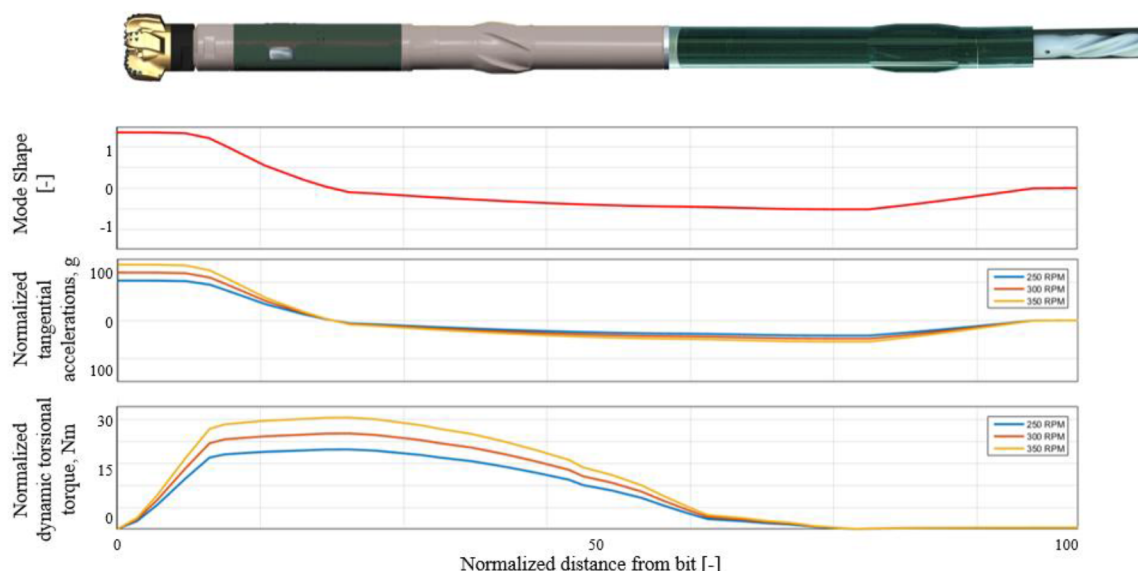


Figure 8—Results of Torsional Oscillation Analyzer (TOA) for actual BHA motorized RSS for 8 1/2" section

TOA uses a stability criterion (Sc) to rank torsional modes by their susceptibility to excitation. All three bit designs tested showed equal Sc values, indicating that any differences in measured torsional stability are due to the bit cutting structure, not the BHA setup.

Table 1—Comparison of BHA setup and susceptibility to torsional excitations

	Design A	Design B	Design C
Bit Length	10.3 in	10.3 in	9.45 in
Steel or Matrix	Matrix	Matrix	Matrix
Number of blades	5	5	6
BHA	Motorized RSS	Motorized RSS	Motorized RSS
Stability criterion	Baseline	Equal to baseline	Equal to baseline

Bit influence

Following the BHA simulations, a 3D computational drilling model was employed to evaluate and optimize drill bit designs. (Matthews et al., 2021). It represents PDC cutters, tricone inserts, and bit bodies using a polygonal mesh that captures detailed geometry. The model includes a physics-based force model, calibrated through full-scale drilling tests under realistic confining pressures. It predicts key performance metrics such as rate of penetration (ROP), steerability, cutter engagement, and contact forces, based on inputs like weight on bit (WOB), RPM, rock type, and pressure. It also compares lateral and torsional stability across bit designs. For bit selection, the model simulates drilling through Alabama marble limestone at 6000 PSI pressure representative for the lab tests and the application, see Fig 9. It captures interactions between PDC cutters, tungsten carbide inserts (TCI), and different rocks, and accounts for rubbing on the bit body.

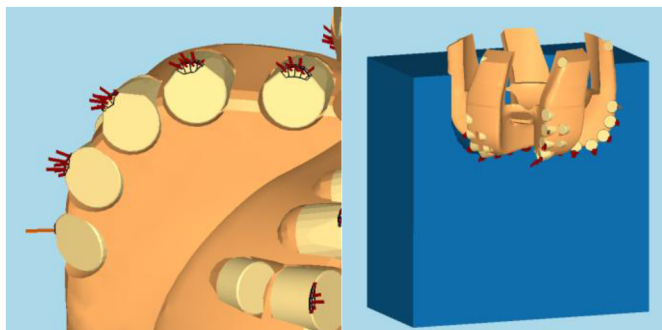


Figure 9—Example of Tetrahedron simulation showing the cutter engagement and cutting forces

The torsional stability of the three bit designs is predicted by simulating the test protocol later used in the laboratory tests. Multiple combinations of WOB and RPM are simulated to generate the cutting forces at every cutter. The cutting forces are responsible for exciting torsional vibrations, specifically high-frequency torsional oscillations. The cutting forces are translated to a torsional stability indicator that is plotted as a color code Fig 10. The plot depicts combinations of DOC over RPM to enable an estimation of the associated ROP. Jumps of the stability index indicate the transition from unstable to stable drilling state. The resulting stability border splits the map into an unstable HFTO window in the top left and into a stable HFTO free window in the bottom right. The size of the stable HFTO free window is an indicator for the torsional stability of the bit designs. As can be seen design C has the largest stability window and is predicted to be more stable than design B. The latter has a larger window than design A and is, hence, predicted to be more stable than design A. These simulations predict the design changes to be effective for mitigating HFTO excitation. Simulations help to reduce the risks of missing design flaws and optimizing the bit design before going into lab and field trials.

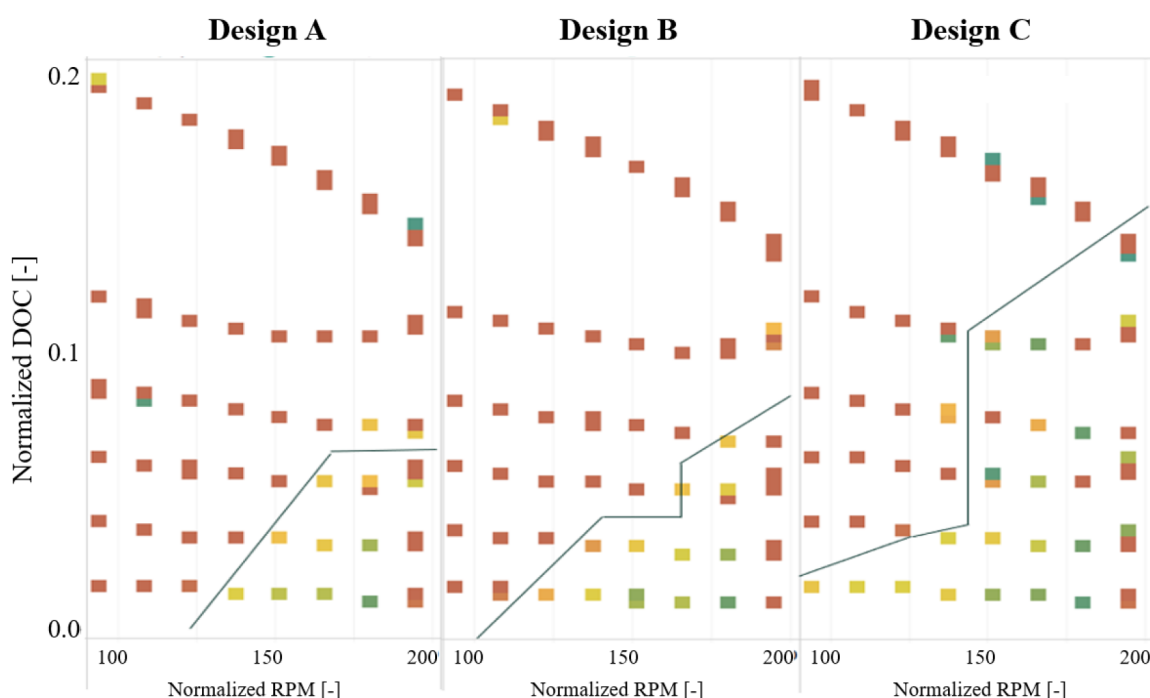


Figure 10—Predicted HFTO stability maps of the three bit designs.

Lab testing

After simulatively predicting the drilling behavior of the three investigated bit designs A, B and C, their performance is validated in a full-scale laboratory to study their behavior in a controllable and observable environment. The lab tests consist of a torsional stability test and lateral stability test. Once certainty about the drilling performance of the three bits is achieved, they are deployed in field operations.

Lab testing capabilities

The full-scale pressurized laboratory drilling rig enables controlled generation of high-frequency torsional oscillations (HFTO), as described by [Everhard et al. \(2023\)](#). The test rig incorporates a pressurized rock chamber and a simplified bottomhole assembly (BHA, [Fig 11](#)), allowing systematic variation of the rate of penetration (ROP), weight on bit (WOB), rotary speed (RPM), bit type, and lithology. The system includes a 200-bbl open-loop mud circulation system, two 1000-hp triplex pumps (maximum 500 gal/min), a 10,000-lbf hoisting mechanism, a 1000-hp rotary drive (up to 10,000 ft-lbf torque), and a pressure vessel rated to 10,000 psi. All tests are operated in an observable and controllable environment ([Kueck et al. 2023](#)).

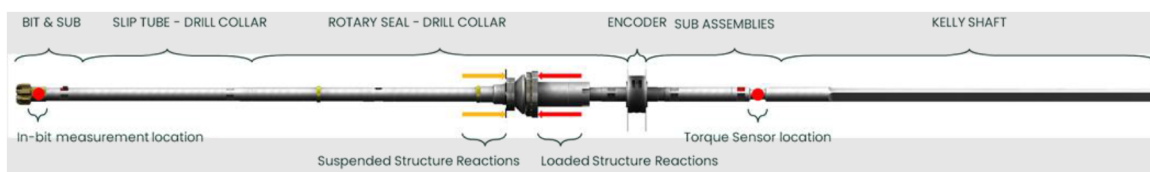


Figure 11—Drill string assembly indicating individual components and locations of in-bit measurement and dynamic torque (red dots)

Dynamic measurements are acquired using a high-frequency in-bit sensing device, see [Fig 12](#) and [Fig 13](#) ([Townsend et al. 2021](#); [Stolboushkin et al. 2024](#)), which integrates tri-axial accelerometers (± 40 g, 2000 Hz), a gyroscope (± 666 rpm, 1.6 kHz), a magnetometer (± 16 gauss), and a 200-g accelerometer for high-frequency vibration analysis. A collocated torque sensor, comprising four strain gauges spaced 90° apart, is coupled with a data acquisition system operating at 5012 Hz.

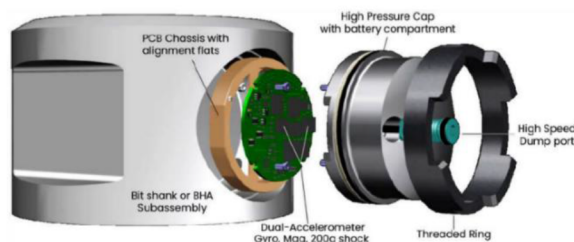


Figure 12—In-bit sensing device with high-frequency gyroscope, accelerometers and magnetometers (Townsend et al. 2021)



Figure 13—Bit sensing device mounted in a bit (circular metallic cap)

Finite element modeling of the rig structure was conducted to identify dominant torsional modes, Fig 14. The second torsional mode, at 141 Hz, was found to be the most susceptible to torsional excitation. The bit location was identified as the optimal position for capturing torsional accelerations associated with these modes.

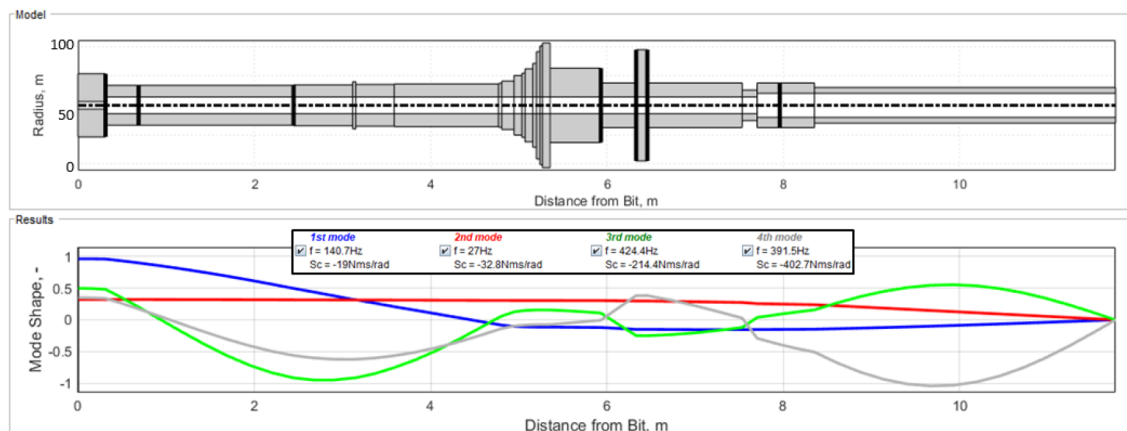


Figure 14—Model of laboratory rig and simulated torsional modes

A standardized testing protocol was implemented to evaluate bit performance and dynamic stability. Tests were conducted on homogeneous carbonate rock under 6,000 psi confining pressure, simulating downhole conditions. A matrix of WOB and RPM combinations was stepwise applied, with each combination held until steady-state drilling was achieved. The resulting data enable quantitative characterization of the bit response and facilitate an extrapolation to field-scale BHAs (Hanafy et al. 2024b).

Torsional stability test

The torsional stability test stepwise applies a range of WOBs and RPMs to evaluate the drilling state and plot a heat map of torsionally instable and torsionally stable operative windows (see also Kueck et al. 2024). Additional drilling characteristics can be deduced such as ROP, MSE, lateral accelerations, aggressiveness to name a few.

Time-based data. Fig 15 shows an example of the deployed testing protocol. The first and second line depict the stepwise increase of the RPM over a constant WOB repeated for multiple WOB levels. The ROP increases with RPM and with WOB as visible in the third row. The fourth row plots the tangential accelerations from the high-frequency sensors in the bit. As can be seen, there are sections of stepwise increase of tangentials and section of low amplitude random tangentials. The stepwise sections are HFTO, as HFTO scale along the increase of RPM. The random sections show zones of mitigated HFTO. The points of transition between stepwise increase and random vibrations mark the threshold of stabilizing RPM that need to be passed to mitigate HFTO. These thresholds increase with increasing WOB making it harder to mitigate HFTO at higher weights. Additionally, the tangential accelerations are used to color the RPM and WOB readings in the first two rows to highlight tangentials scale with RPM.

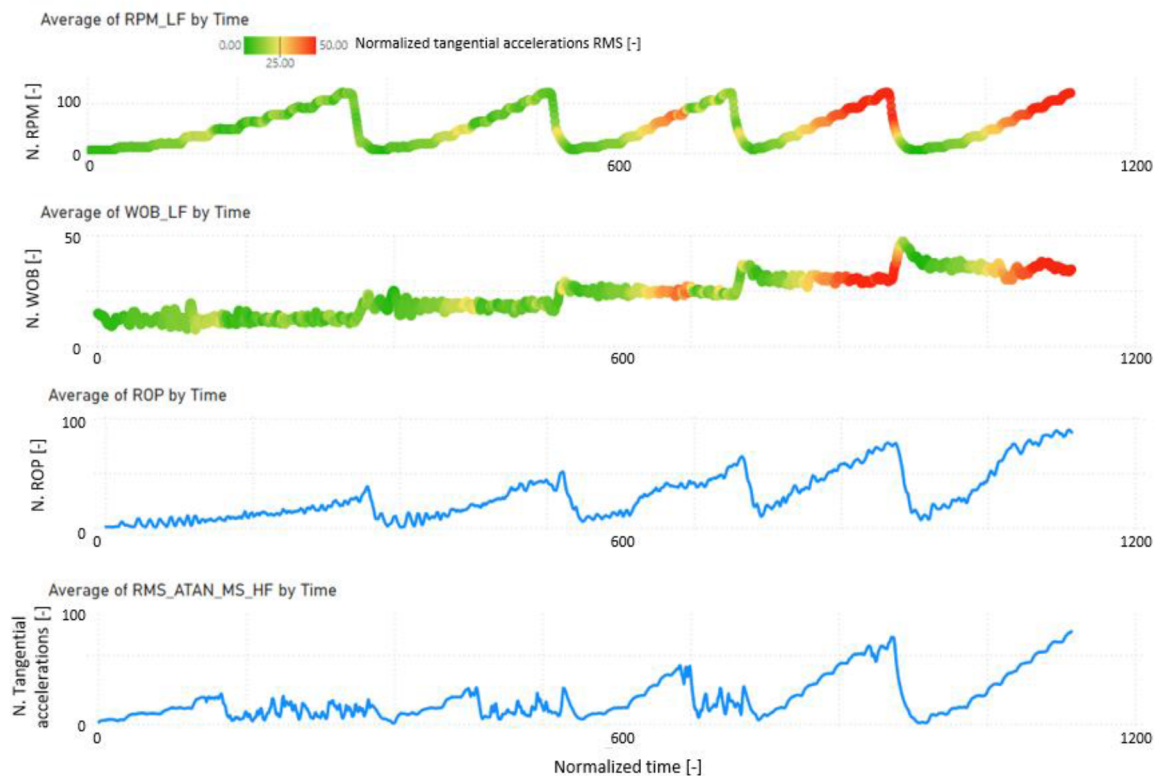


Figure 15—Example test protocol for tested bits in the lab under pressure.

Steady bit response. The stepwise test protocol is applied on all three bit designs A, B and C. In a first assessment, the data are statistically evaluated to compare the median and the range of performance values. The median values enable to directly compare the performance while the distribution enables to compare the variation and peak values of the performance. Fig 16 shows the ROP, MSE and Mu as defined by the ratio of torque to WOB. As can be seen, changes to design B increase the median ROP by about 15% compared to design A. Bit design C achieves a higher peak ROP, however the median ROP is in-between designs A and B, meaning there is a great variation of ROP depending on the applied WOB and RPMs. Looking at the MSE as a measure for how efficiently energy is translated into removal of a volume of rock, design B drills slightly more efficiently than design A as the median and peak MSE are lower compared to design A. Design C shows a different behavior as the MSE is spread out indicating a high dependency of the drilling efficiency on the operative parameters. The median MSE is higher than in designs A and B pointing towards an increased occurrence of inefficient drilling states overall. The bit aggressiveness measured by the ratio of torque to weight is lowest in design A and increases towards designs B and C. While design B is able to translate the increased torque per weight on bit to a higher ROP also shown by the higher drilling efficiency,

design C generates more torque per weight on bit not turning into higher ROP also indicated by the higher MSE as a measure for inefficient drilling.

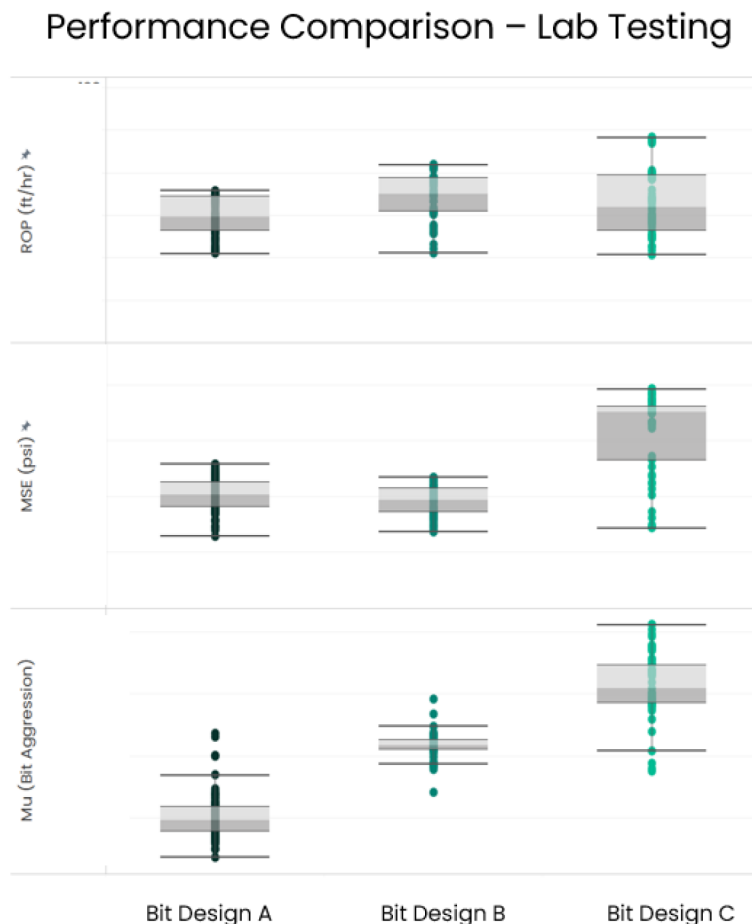


Figure 16—Comparison of ROP, MSE and Mu

Torsional stability. The time based data are transformed to stability maps plotting WOB over RPM. Every combination of the two operative parameters is color-coded by the normalized tangential accelerations as an indicator for the existence of HFTO at this point. The resulting map is split in two zones, an HFTO zone at high WOB and low RPM and an HFTO-free zone or torsional stability window at low WOB and high RPM. The size of the torsional stability window is an measure for the torsional stability of a bit. As depicted in Fig 17, the bit designs A, B and C possess different torsional stabilities showing that the applied design changes have an influence on the torsional excitation of the bit. Particularly, design C has a very large HFTO free window (83% no HFTO) and is almost HFTO free in the chosen range of operative parameters. Design B shows improvements over design A, increasing from 16% to 53% HFTO free window. Design C shows an five times larger torsional stability window than design A. The measured results correspond well to the simulative prediction of the torsional stability. Designs B and C are faster than design A, therefore, it can be stated that higher ROP and higher torsional stability simultaneously is possible and does not require compromises.

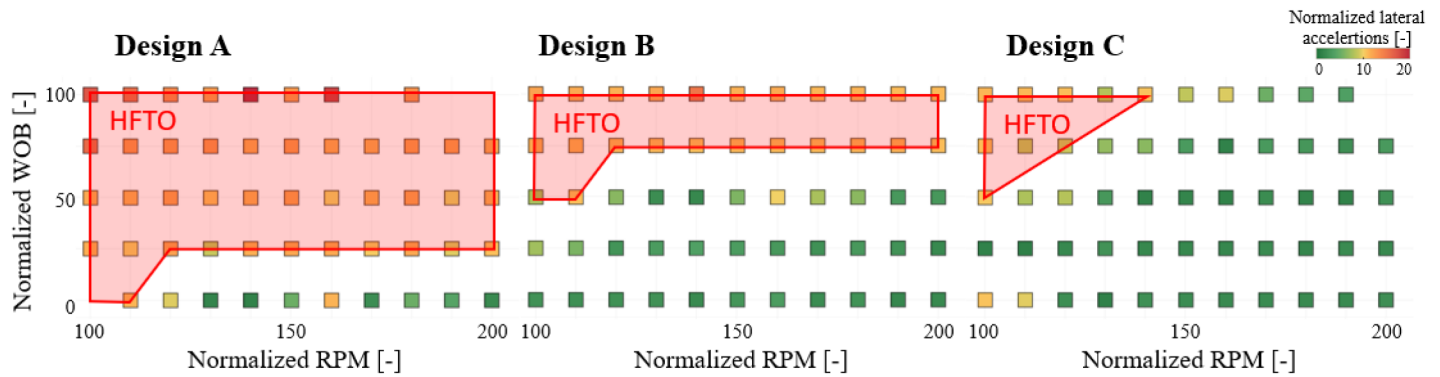


Figure 17—Comparison of torsional stability maps

Lateral stability test

To qualify the lateral stability of the three bit designs, they were deployed in an atmospheric pressure test increasing the WOB stepwise while keeping the RPM constant. This enables to assess the lateral stability through different rate-of-penetrations and depth-of-cuts. One indicator for the lateral stability used in this study is the hole diameter generated at different DOCs, Fig 18. As can be seen in the plot, Designs A and B show a very similar steady hole size over different DOCs, design B might show a little higher hole oversize at low DOCs. Design C on the other hand produces a significantly larger hole diameter at low DOC converging to the same stable holes size as designs A and B.

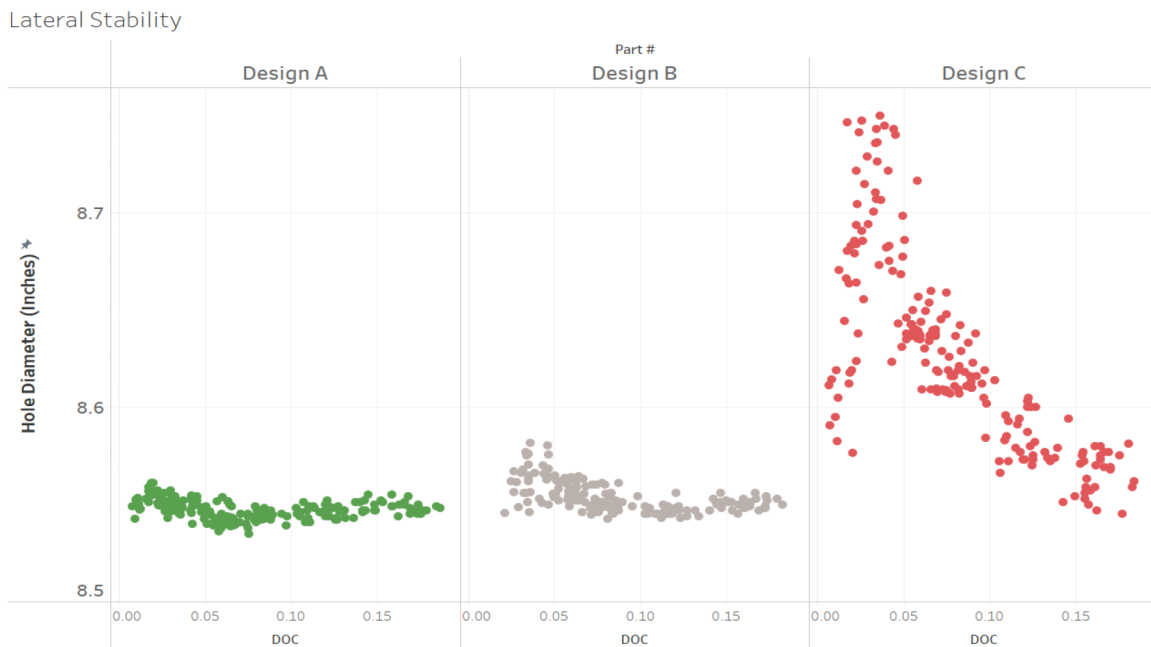


Figure 18—Hole diameter comparison from atmospheric testing.
Design A is at far left, Design B at center, and Design C at right.

This can also be observed when comparing the bottomhole patterns generated by the moving cutting structure documenting the end of the test in Fig 19, Fig 20 and Fig 21. Designs A and B produce nice circular cutting grooves indicating a stable concentric drilling state. Design C shows a slight polygonal shape proof of additional lateral movement that was more pronounced in the beginning of the test at lower DOCs.



Figure 19—Bottom Hole Pattern Design A

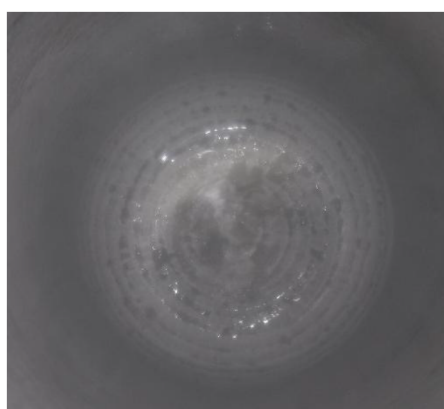


Figure 20—Bottom Hole Pattern Design B



Figure 21—Bottom Hole Pattern Design C

Summary lab testing

The lab tests show that design B is slightly faster and more aggressive than design A. It also shows a higher drilling efficiency, a higher torsional stability and a comparable lateral stability. Design C shows the highest peak ROP, but lower median ROP, the drilling efficiency is lower than designs A and B, while the aggressiveness is highest, which does not translate into higher ROP. The torsional stability is highest of all designs, whereas the lateral stability is lowest. Overall, design B portrays a good evolution of design A improving all aspects, whereas design C is a more radical design change with highest torsional stability and peak ROP, but lower lateral stability and lower drilling efficiency showing a higher risk-reward ratio.

Field testing

Overview

A field area in the Middle East was selected for testing. An 8½" PDC bit and a 6¾" motorized RSS BHA were used across all offset wells, which had similar trajectories. The objective was to land the well from vertical to horizontal while drilling through the formations mentioned earlier.

Two approaches were used to evaluate the test results:

1. Parameter-Step Tests: Conducted under parameters similar to those used in the lab.
2. Vibration Time Analysis: Evaluated the total duration of high-frequency tangential oscillations and lateral vibrations relative to the total drilling time for each run.

Parameter-step tests and stability maps

During the field parameter-step tests, surface RPM varied widely - from 20 to 110 RPM - with multiple WOB (Weight on Bit) steps, as shown in the figure below, to identify the stability boundary. A stable flow rate was maintained throughout the tests to replicate lab conditions. Tests were conducted in Formations II and III, as both are more prone to various types of vibrations, particularly HFTO. Formation IV was excluded due to its highly interbedded composition of anhydrite and limestone, which complicates vibration analysis. [Figures 22 and 23](#) show the planned test parameters versus the actual values recorded during the tests. Please note that these step tests are usually used to identify parameter combinations of low MSE ([Watson et al. 2023](#), [Samuel et. al. 2022](#)), however, in this study tangential accelerations are the target parameter.

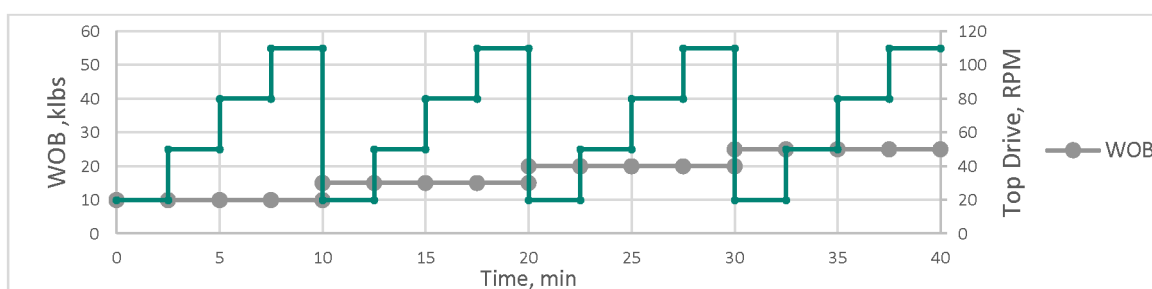


Figure 22—snapshot for Planned Parameter-Step test Performance (WOB and SRPM)



Figure 23—snapshot for actual Parameter-Step test Performance (Actual WOB and total bit RPM including downhole motor)

High-frequency vibration data were retrieved from the RSS memory for additional validation. A snapshot of the data is shown in [Figure 24](#), highlighting when HFTO was triggered and when it was not. The measured HFTO frequencies aligned with the modeled predictions. Based on the modeling and actual frequency data, the bit inertia can also be estimated.

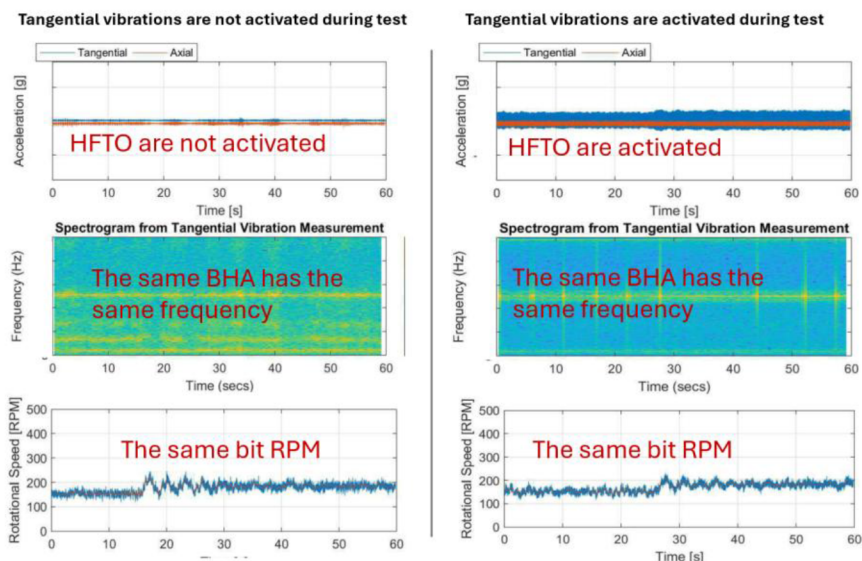


Figure 24—High Frequency Memory data during parameter-step test

Torsional stability maps based on parameter-step test

Field time based data are transformed to stability maps plotting WOB over RPM for every parameter-step test to compare with the lab results. The resulting map shows similar trend to the lab and shows the split in two zones, an HFTO zone at high WOB and low RPM and an HFTO-free zone or torsional stability window at low WOB and high RPM. The size of the torsional stability window is an measure for the torsional stability of a bit.

Design A vs Design B in Formation II (inside limestone)

Field parameter-step tests in the formation II have shown the Design B improved HFTO free window by approximately +25% over base design as depicted in Fig 17 and Fig 26. Design B have shown fully HFTO free window at first WOB step. There was no parameter-step test of Design C in the Formation II. These field results confirm the lab results and allows for further optimizations.

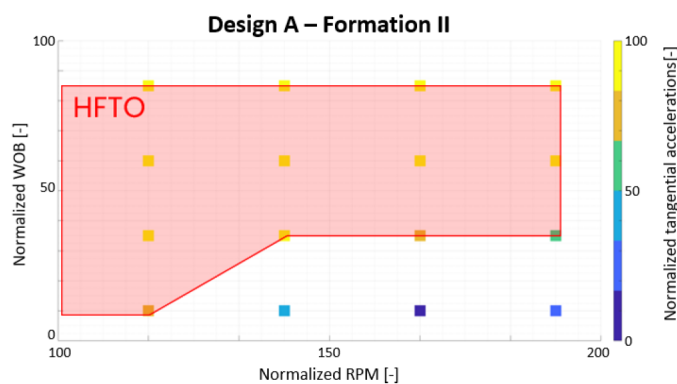


Figure 25—HFTO Stability window with design A

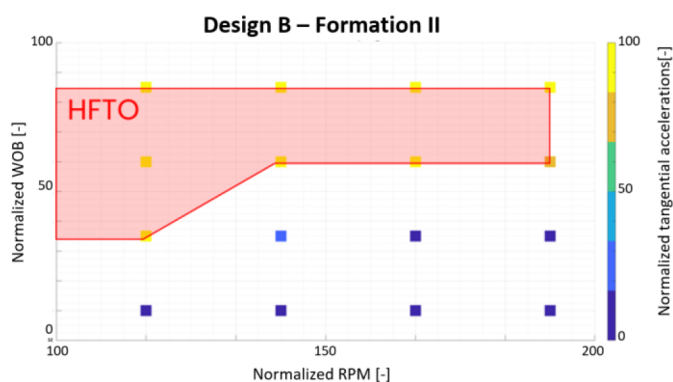


Figure 26—HFTO Stability window with design B

Design B vs Design C in Formation III

Parameter-Step tests in the formation III (anhydrite) have shown the Design B has very narrow free window. Generally it was expected in this formation to have narrow window due to high UCS and generally field observation. It was interesting that design C significantly improved HFTO free window by approximately +30% and even showed the highest HFTO free window for all parameter-step field test as depicted in Fig 17 and Fig 28. It showed good potential for future improvement while drilling similar formation. The design frame will be studied further to be improved.

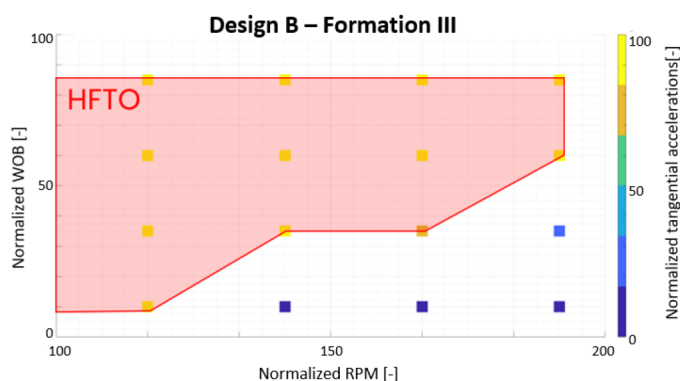


Figure 27—HFTO Stability window with design B

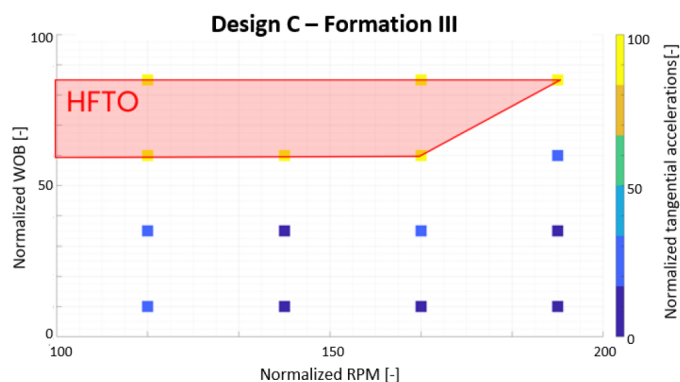


Figure 28—HFTO Stability window with design C

Torsional stability based on averaged data

Designs A, B, and C were evaluated across multiple wells in formations II and III to quantify High-Frequency Torsional Oscillations (HFTO) as a percentage of on-bottom drilling time. HFTO classification

was based on Bottom Hole Assembly (BHA) analysis, using a normalized threshold for tangential vibration levels to differentiate between the 'No HFTO' (green) and 'HFTO' (red) categories. The accompanying figure illustrates the performance progression across the different designs, showing how Designs B and C have progressively expanded the HFTO free drilling time from 19% with design A to 25% with design B to 37% time drilled without HFTO with design C. This trend is consistent with results from both simulation and laboratory testing each showing the same order of torsional stability. The observed correlation between simulations, lab experiments, and field data supports more informed design iterations aimed at further enhancing HFTO stability.

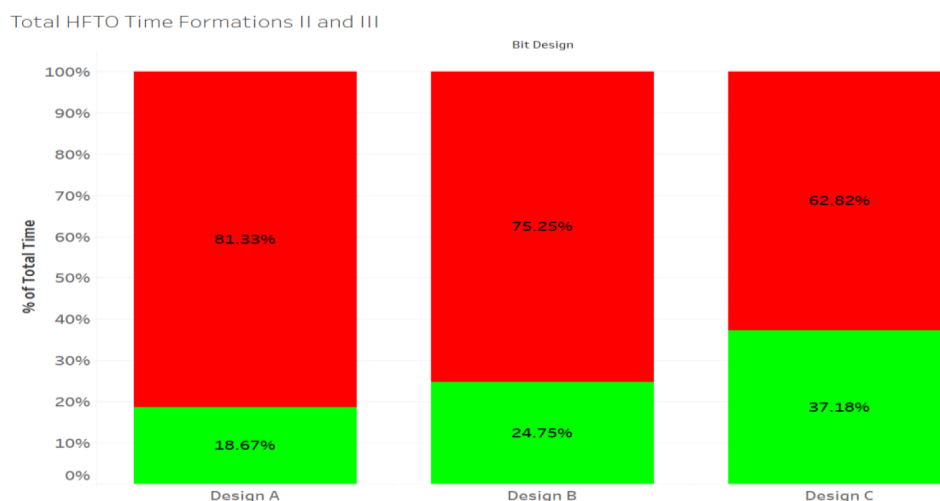


Figure 29—"HFTO%" in red and "NO HFTO %" in Green

Lateral stability based on averaged data

Designs A, B, and C were also evaluated in the same wells in formations II and III to assess lateral vibrations, considering both the magnitude and the percentage of on-bottom drilling time, as well as insights derived from parameter-based stability maps. Unlike the HFTO analysis, no fixed threshold was applied; instead, lateral vibrations were quantified across a range of acceleration magnitudes (g-levels) to better understand their distribution and the operational conditions under which they occur. Figure 30 illustrates that lateral vibration levels increased with Designs B and C compared to Design A, aligning with the lateral stability rankings observed during laboratory testing. However, the lateral stability map in Fig 31 reveals that, particularly for Design C, elevated lateral vibrations are concentrated in regions characterized by high RPM and high Weight on Bit (WOB), corresponding to high Depth of Cut (DOC) and high Rate of Penetration (ROP) conditions. This contrasts with laboratory observations, where lateral vibrations predominantly occurred under low WOB, high RPM, and low DOC conditions. While the consistency in design ranking between lab and field tests is encouraging, further investigation is required to understand the discrepancy in the DOC environments associated with lateral vibration onset in lab versus field conditions. One of the explanations could be that field RSS BHA is more flexible and bend more than lab BHA at high WOB. It led to elevated vibrations at high WOB compared to the Lab.

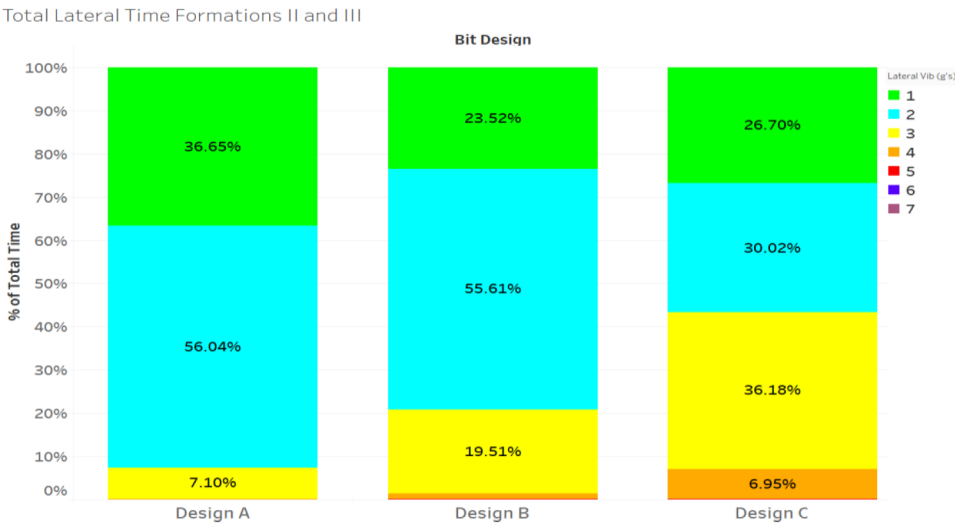


Figure 30—Lateral vibration % per range of acceleration magnitudes (g-levels)

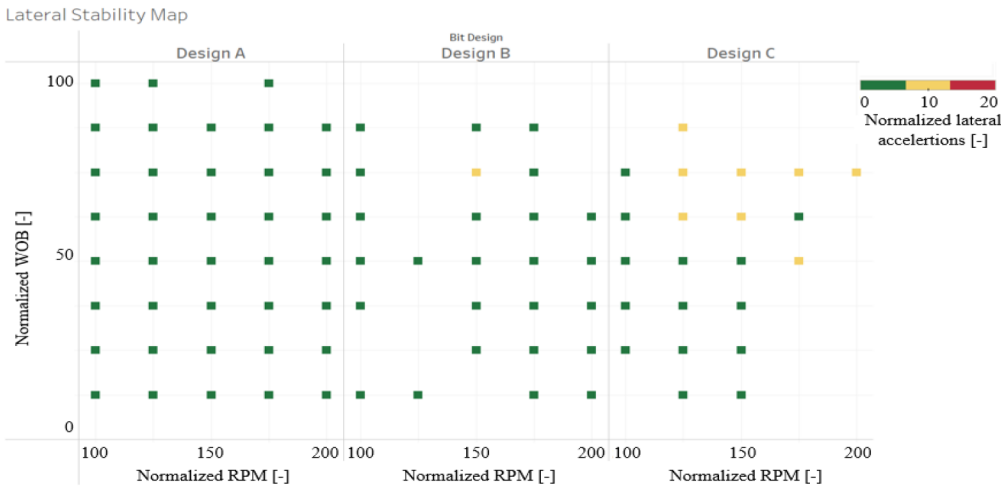


Figure 31—Parameter-based stability maps colored by Lateral Vibration Levels

Summary comparison

3 bits designs were rated with scale 1 to 3. Where 1 was the best out of 3 bits. If both are similar, they were equal. Torsional and lateral stability were matching lab results and field results. Field results also have shown to us that both are important and any of them can be bottleneck leading to tool failures or controlled ROP. To maximize field performance for future design we need to address torsional and lateral stability separately. Design C durability issue might happen due to higher lateral instability, and it could lead to lower ROP results than lab results. Also design C had lowest score in Lab MSE which could be a reason for slightly lower field performance, see [table 2](#). Still design C showed the highest torsional stability against HFTO in the lab and in the field. The future PDC design ideas will include main features from this design after addressing lateral stability and ROP aspects.

Table 2—Overview of simulation, lab and field results

Design	Lab Data					Field Data			
	ROP	MSE	MU	torsional stability	lateral stability	ROP	torsional stability	lateral stability	Durability
A	3	2	3	3	1	1	3	1	1
B	2	1	2	2	1	1	2	2	1
C	1	3	1	1	2	2	1	3	2

Summary

- This study reports one of the first results of stability maps that are generated on the rig in field conditions. The trends agree with lab results providing confidence in the improvements and understanding generated based on laboratory measurements. As a second measure independent to stability maps, the percent time in HFTO and lateral vibration is also determined.
- A new simulative approach in developing PDC drill bits is introduced and validated with lab and actual field results. It demonstrates the correct calibration of the simulation computational model, and the understanding of how to transfer simulations to observable field outcomes.
- Three drill bit designs were compared in terms of drilling performance and vibration. It was shown that design B and design C progressively increased the stability window by 37% and 67% respectively in terms of torsional stability over the baseline design A in the lab, which also translated into increases of the field stability windows of 25% and 30% respectively. Even though Design C achieved the highest torsional stability, it also showed the weakest lateral stability. Design B landed at a good balance of torsional and lateral stability and ROP.
- The presented steps are essential for future design improvements in critical applications where high lateral vibrations and HFTO are present. Torsional stability, lateral stability and ROP will be addressed without compromise to maximize the field performance of future designs.

Acknowledgements

The authors thank Baker Hughes for the support and the permission to publish this research. Thanks to the Woodlands Technology Center Lab Team, Kai Glowatzki, Mohamed Benzaoui and Eric Fankhauser for supporting the lab tests and the data analyses in this paper.

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